

PAPER Vacuum-Vessel and Support Structural Mechanics Under Electromagnetic, Halo-Current, and Seismic Loads for a Compact Spherical-Tokamak Breeder

P I Ford¹, R J Weggel¹ and C Weggel¹ ¹Kronos Fusion Energy, Los Angeles, California, USA

ORCID (P I Ford): 0000-0003-0395-1752

Abstract

The vacuum vessel and its support carry the entire electromagnetic, thermal and inertial load path of a compact spherical-tokamak breeder into the ground, and in a high-power-density device the impulsive disruption load, not the steady field or the seismic base excitation, sets the design. We formalise the vessel-and-support structural-mechanics problem for the Hyperion breeder (design point $I_p = 9.66$ MA, $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, negative triangularity $\delta = -0.30$, $B_0 = 8$ T, elongation $\kappa = 2.0$, major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, centrepost peak field 16.84 T) as a coupled thermo-electro-mechanical continuum problem, and we solve it load case by load case with retained margin. We write the governing momentum balance, the Maxwell-stress electromagnetic body force, the thermoelastic constitutive law, the halo-current and eddy force integrals, the modal response-spectrum seismic formulation, and the linear-elastic-fracture-mechanics flaw criterion as explicit equations, and we cast the qualification as a ladder of inequalities against the 800 MPa structural allowable. Under the electromagnetic self-load the 16.84 T field produces a 113 MPa magnetic pressure and a near-hydrostatic -190 MPa compression that holds the equivalent von Mises stress to 84 MPa; a combined cooldown-plus-quench-plus-electromagnetic transient with fast quench detection caps the hot-spot at 63 K and the combined von Mises at 245 MPa ($3.3\times$), so detection speed *is* structural margin; the highest-fidelity finite-element model on the real centrepost CAD keeps the structural jacket to 579 MPa ($1.38\times$) and the conductor hoop strain to 0.151% ($2.65\times$), and an engineered per-turn precompression winding cuts the peak REBCO hoop strain by 52.4% and flattens its profile $4.8\times$. The disruption halo/eddy envelope — halo current ≈ 2.9 MA, vertical force ≈ 23 MN over a 7–19 ms current quench — dominates the support design, standing two orders of magnitude above a design-basis seismic base shear of ≈ 0.2 MN; the cryogenic 316LN jacket is leak-before-break, with a critical flaw of 333 mm at operating stress against a 2 mm inspection threshold and a Paris-law fatigue life of order 10^8 cycles. Every result is grounded in the Kronos Physics De-Risking Register and was produced on a GPU-accelerated finite-element campaign. The binding constraint on the structure is neutron lifetime, not stress; the burner plug coil at 26.49 T is a distinct magnet, analysed separately.

Keywords: structural mechanics; vacuum vessel; disruption halo current; seismic response spectrum; leak-before-break; REBCO magnet; spherical-tokamak breeder

1 Introduction

1.1 The structural problem of a compact spherical-tokamak breeder

A spherical tokamak (ST) confines a large plasma current in a small volume, which is what makes it an attractive neutron source and breeder platform [1, 2] and, at the same time, what concentrates its mechanical loads. The Hyperion breeder holds $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m behind a centrepost magnet at a peak field of 16.84 T, in a negative-triangularity ($\delta = -0.30$) regime that accesses good confinement without the edge pressure pedestal and the associated edge-localised-mode cycle of conventional H-mode [3, 4]. The vacuum vessel is the pressure and vacuum boundary, the first structural ring outside the plasma, and the member that carries every load the machine generates — the steady electromagnetic self-load of the magnets, the thermal excursions of cooldown and of a quench, the impulsive forces of a plasma disruption, and

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the seismic base excitation of the site — into the support and the ground, while a 14 MeV neutron flux slowly consumes the material of the innermost members.

Three features make this problem distinct from that of a conventional aspect-ratio tokamak. First, the compactness that gives the ST its high power density leaves little radial room for structure precisely where the field and the current density are highest, so the vessel and its support cannot be sized by strength margin alone — they must be reconciled with a tight radial budget. Second, the high plasma current in a small volume makes the disruption the governing mechanical event: at $I_p = 9.66$ MA a vertical displacement event drives halo currents of mega-ampere scale through the vessel and the support in milliseconds, and the resulting Lorentz forces dwarf the steady loads. Third, the small seismic mass of a compact machine makes the seismic base shear small in absolute terms, so that the seismic case — which sizes the support of a large light-water reactor — is here a subdominant check rather than the governing load. Quantifying that hierarchy, and closing every load case against a single allowable, is the purpose of this paper. Figure 1 orients the load path: the electromagnetic self-load, the disruption halo force, and the seismic base excitation all resolve through the vacuum vessel and the bucked-and-wedged support into the ground.

Vacuum vessel, support, and centrepost load path (Schematic.)

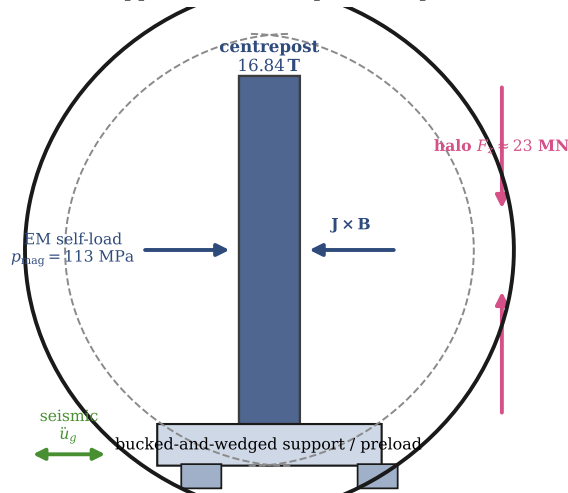


Figure 1. (Schematic.) The vacuum vessel, bucked-and-wedged support, and 16.84 T centrepost, and the three load classes they carry: the electromagnetic self-load $\mathbf{J} \times \mathbf{B}$, the disruption halo vertical force $F_z \approx 23$ MN, and the seismic base excitation $\ddot{u}_g$. The figure encodes the load path only; no data values are plotted.

1.2 Prior art

Tokamak vessel and support structural mechanics is a mature engineering discipline. The ITER vacuum vessel is a double-walled, code-qualified structure whose design and construction have been documented in detail [5], and the disruption electromagnetic-load basis for large tokamaks is codified in the ITER Physics Basis [6] and in the disruption-mitigation strategy that follows from it [7, 8]. Halo currents — the poloidal currents that flow through the plasma scrape-off layer and close through the conducting vessel during a vertical displacement event — have been characterised experimentally, notably on the National Spherical Torus Experiment, where their toroidal asymmetry and rotation set the local force peaking [9], and the vertical forces they generate in an ITER-scale plasma have been quantified with three-dimensional magneto-hydrodynamic modelling [10]. The eddy-current and passive-conductor problem that accompanies the halo load is now addressed with dedicated thin-wall electromagnetic codes [11]. On the magnet side, the structural design of a spherical-tokamak centre stack under combined electromagnetic, thermal and structural loading was worked out in detail for the NSTX-Upgrade ohmic-heating coil and centre stack [12], and the cryogenic structural materials that carry these loads — austenitic stainless steels at 4 K — have an established fracture-toughness and fatigue database. The mechanics of the REBCO coated conductor that the field acts on, including its strain tolerance, is likewise well characterised [13].

What the literature does not yet provide, for a compact high-field ST breeder, is a single continuum formulation that (i) writes the electromagnetic, thermal, halo/eddy and seismic loads on the vessel and support as a coupled thermo-electro-mechanical problem with the load cases

superposed against one allowable, (ii) ranks those load cases quantitatively so that the governing case is identified rather than assumed, and (iii) closes the fracture-mechanics argument that makes the cryogenic structure leak-before-break rather than merely below yield. This paper contributes that formulation and its solution.

1.3 Contribution

We give a formal statement of the vessel-and-support structural problem as a coupled continuum boundary-value problem (§2), deriving the electromagnetic body force from the Maxwell stress tensor, the thermoelastic stress from the constrained-cooldown-plus-quench temperature field, the halo/eddy force envelope from the disruption current path, the modal seismic response from a design response spectrum, and the flaw criterion from linear-elastic fracture mechanics. We describe the GPU-accelerated finite-element campaign that produced the results, naming the codes actually run and the verification and reproducibility protocol (§3). We then present the qualification load case by load case (§4): the electromagnetic self-load, the load-case ladder with detection speed as margin, the engineered precompression winding, the disruption halo/eddy envelope, the design-basis seismic response, the governing-load hierarchy, the leak-before-break fracture argument, and the retained-margin summary. We compare each result quantitatively to the published tokamak and cryogenic-structures literature (§5), state the one open confirming calculation (§6), and conclude that the structure is a solved problem whose binding constraint is neutron lifetime. Every quantitative claim traces to a frozen result in the Kronos Physics De-Risking Register [30]; the serial identifiers (e.g. QS-14, P3-FEA-01) are given inline so that each number is auditable. The breeder centrepost at 16.84 T and the burner plug coil at 26.49 T are treated throughout as the two distinct magnets they are, and are never conflated.

2 Governing equations and the formal problem

2.1 Continuum balance and thermoelastic constitution

The vessel and support are treated as a thermoelastic continuum occupying a domain Ω with boundary $\partial\Omega$. Writing $\mathbf{u}(\mathbf{x}, t)$ for the displacement field, $\boldsymbol{\sigma}$ for the Cauchy stress and ρ_m for the mass density, the balance of linear momentum is

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{f}_b = \rho_m \frac{\partial^2 \mathbf{u}}{\partial t^2}, \quad (1)$$

with $\mathbf{f}_b$ the body-force density. For the steady and quasi-static load cases the inertial term is dropped and (1) reduces to $\nabla \cdot \boldsymbol{\sigma} + \mathbf{f}_b = \mathbf{0}$; it is retained in full for the seismic case of §2.6. The kinematics are linear (small strain), so the strain tensor is the symmetric gradient

$$\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top). \quad (2)$$

The material is isotropic and thermoelastic, so the constitutive law couples mechanical strain and a temperature excursion $\Delta T = T - T_{\text{ref}}$ through the linear-thermal-expansion coefficient α ,

$$\boldsymbol{\sigma} = 2\mu \boldsymbol{\varepsilon} + \lambda \text{tr}(\boldsymbol{\varepsilon}) \mathbf{I} - (3\lambda + 2\mu) \alpha \Delta T \mathbf{I}, \quad (3)$$

with Lamé parameters $\mu = E/[2(1 + \nu)]$ and $\lambda = E\nu/[(1 + \nu)(1 - 2\nu)]$ in terms of Young's modulus E and Poisson's ratio ν . The scalar quantity that drives yielding is the von Mises equivalent stress, built from the deviatoric stress $\mathbf{s} = \boldsymbol{\sigma} - \frac{1}{3}\text{tr}(\boldsymbol{\sigma}) \mathbf{I}$,

$$\sigma_{\text{vM}} = \sqrt{\frac{3}{2} \mathbf{s} : \mathbf{s}} = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]}, \quad (4)$$

with $\sigma_1, \sigma_2, \sigma_3$ the principal stresses. Qualification requires $\sigma_{\text{vM}} \leq \sigma_{\text{allow}}$ with $\sigma_{\text{allow}} = 800$ MPa for the steel-fraction structural members analysed here (QS-09). Equation (4) makes explicit why a near-hydrostatic state is benign: when $\sigma_1 \approx \sigma_2 \approx \sigma_3$ the deviator vanishes and $\sigma_{\text{vM}} \rightarrow 0$ regardless of the pressure magnitude.

2.2 Electromagnetic body force and the Maxwell stress

The steadiest load is the magnet field acting on its own current. In the magnetoquasistatic limit the body force is the Lorentz density

$$\mathbf{f}_b = \mathbf{J} \times \mathbf{B}, \quad (5)$$

which, using Ampère's law $\mu_0 \mathbf{J} = \nabla \times \mathbf{B}$ and a vector identity, is the divergence of the Maxwell stress tensor

$$T_{ij} = \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right), \quad \mathbf{f}_b = \nabla \cdot \mathbf{T}. \quad (6)$$

The normal traction that (6) exerts across a surface tangent to the field is the magnetic pressure

$$p_{\text{mag}} = \frac{B^2}{2\mu_0}, \quad (7)$$

which at the centrepost peak field evaluates to

$$p_{\text{mag}}(16.84 \text{ T}) = \frac{(16.84)^2}{2\mu_0} = 113 \text{ MPa} \quad (\text{QS-09}). \quad (8)$$

Because the centrepost is held in compression by a bucked-and-wedged support (§2.4), the resulting stress state is near-hydrostatic — a -190 MPa compression — so by (4) the equivalent stress is only $\sigma_{\text{vM}} = 84 \text{ MPa}$, far below σ_{allow} . The electromagnetic self-load is not the binding case; it establishes the compressive baseline on which the thermal and disruption cases superpose. The burner plug coil, a distinct magnet, reaches $p_{\text{mag}}(26.49 \text{ T}) = 279 \text{ MPa}$ by the same equation (7) and is qualified separately (§4.8).

2.3 Thermoelastic stress and the detection-speed coupling

The temperature excursion ΔT in (3) has two sources: the steady cooldown from ambient to the 4.5 K operating point, and the transient hot-spot of a quench. For a fully constrained member the thermoelastic stress magnitude is

$$\sigma_{\text{th}} = \frac{E \alpha \Delta T}{1 - \nu}, \quad (9)$$

so the quench contribution scales linearly with the hot-spot temperature rise $\Delta T_{\text{hs}} = T_{\text{hs}} - T_{\text{op}}$. The hot-spot temperature is set by the adiabatic heat balance during the detection-and-dump delay: neglecting conduction over the delay, the local current density J heats the conductor according to the “MIITs” integral

$$\int_0^{\tau_{\text{det}} + \tau_{\text{dump}}} J^2(t) dt = \int_{T_{\text{op}}}^{T_{\text{hs}}} \frac{C_v(T)}{\rho_e(T)} dT, \quad (10)$$

with C_v the volumetric heat capacity and ρ_e the electrical resistivity of the normal zone. Equation (10) is monotone in the delay, so a faster detection τ_{det} produces a lower T_{hs} , a lower σ_{th} by (9), and a lower combined von Mises by superposition on the electromagnetic baseline. Concretely (QS-11): an unmitigated quench that lets the hot-spot reach 200 K produces a 550 MPa thermal stress and drives the combined von Mises to 476 MPa ($1.7\times$), whereas fast detection that caps the hot-spot at 63 K holds the combined von Mises to 245 MPa ($3.3\times$). Detection speed is therefore not a diagnostic nicety but a direct structural margin, and it ties the structural case to the demonstrated quench-precursor detection of the magnet twin.

2.4 Support, preload and the demountable joint

The compressive baseline that keeps (4) small is not free; it is maintained by an engineered preload. The bucked-and-wedged support carries a demountable joint whose electrical resistance is contact-pressure-limited. Requiring a joint resistivity $\rho_c = 11 \text{ n}\Omega \text{ cm}^2$ sets a minimum contact pressure

$$p_c \gtrsim 91 \text{ MPa} \iff F_{\text{clamp}} \gtrsim 93 \text{ MN}, \quad (11)$$

and this preload exceeds the vertical-share stress of $\approx 83 \text{ MPa}$ that the joint must react, so the joint never gaps and the electromagnetic load stays compressive through the entire thermal and disruption cycle (QS-13). In the negative-triangularity configuration a Weggel-flared centrepost relieves the ends — the magnetic pressure (7) at the flared end is $\approx 42\%$ lower at the same current — while the throat governs at 81 MPa and is unchanged if preserved; the ohmic-solenoid stack (251 MPa hoop) bucks outward against the toroidal-field set, and the two are mutually relieving (QS-15).

At the conductor scale the Lorentz strain is flattened by an engineered per-turn precompression winding. Grading the preturn precompression redistributes the hoop strain that (5) drives into the no-insulation REBCO pack, cutting the peak REBCO hoop strain by 52.4% (from 0.595% to 0.283%) and reducing the strain-profile peaking factor by $4.8\times$ (SC-BR-4). On the real centrepost CAD this graded preload delivers a finite-element peak conductor hoop strain of 0.151% at a $2.65\times$ margin and a structural-jacket von Mises of 579 MPa at $1.38\times$ (P3-FEA-01), with the canonical magnet dimensions unchanged.

2.5 Disruption halo and eddy forces

The impulsive load is a plasma disruption. During a vertical displacement event the plasma current partially converts to a halo current $I_{\text{halo}} = f_h I_p$ that flows through the scrape-off layer and closes through the conducting vessel and support; the halo fraction f_h and the toroidal peaking factor together set the local force. At $I_p = 9.66$ MA the envelope halo current is $I_{\text{halo}} \approx 2.9$ MA ($f_h \approx 0.30$). The net force on a conducting member is the volume integral of the Lorentz density (5) over the halo path,

$$\mathbf{F} = \int_V \mathbf{J}_{\text{halo}} \times \mathbf{B} \, dV, \quad (12)$$

whose vertical component — the halo current crossing the toroidal field — sets the design envelope. A second contribution arises from the eddy currents induced by the collapsing poloidal field during the current quench. Faraday's law $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ drives a wall current density $\mathbf{J}_{\text{eddy}} = \sigma_e \mathbf{E}$ in the conducting shell of conductivity σ_e , and the induced force scales with the quench rate $\partial \mathbf{B} / \partial t \sim B_p / \tau_{\text{CQ}}$. For a current-quench time in the window $\tau_{\text{CQ}} = 7\text{--}19$ ms the reduced electromagnetic force envelope on the centrepost and vessel is (QS-14)

$$F_z \approx 23 \text{ MN (vertical)}, \quad F_{\text{sw}} \approx 19 \text{ MN (sideways)}, \quad F_{\text{eddy}} \approx 16 \text{ MN}, \quad (13)$$

with the vertical component setting the ≈ 23 MN design envelope. This envelope is carried by a conductive shell with a defined halo-current path and disruption mitigation, keeping the impulsive load within the compressive structural regime of §2.2.

2.6 Seismic base excitation and the modal response spectrum

The seismic load is a base excitation $\ddot{u}_g(t)$ applied through the support. Discretising (1) in space gives the structural equation of motion for the nodal displacement vector $\mathbf{u}$,

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = -\mathbf{M}\mathbf{r} \ddot{u}_g(t), \quad (14)$$

with mass, damping and stiffness matrices $\mathbf{M}, \mathbf{C}, \mathbf{K}$ and influence vector $\mathbf{r}$. The free-vibration modes (ω_n, ϕ_n) solve the generalised eigenproblem

$$(\mathbf{K} - \omega_n^2 \mathbf{M}) \phi_n = \mathbf{0}, \quad (15)$$

and expanding $\mathbf{u} = \sum_n \phi_n q_n(t)$ decouples (14) into modal oscillators

$$\ddot{q}_n + 2\zeta_n \omega_n \dot{q}_n + \omega_n^2 q_n = -\Gamma_n \ddot{u}_g(t), \quad \Gamma_n = \frac{\phi_n^\top \mathbf{M} \mathbf{r}}{\phi_n^\top \mathbf{M} \phi_n}, \quad (16)$$

with modal participation factor Γ_n and damping ratio ζ_n . The peak modal displacement is read directly from the design response spectrum $S_a(\omega_n, \zeta_n)$ as $q_{n,\text{max}} = \Gamma_n S_a(\omega_n, \zeta_n) / \omega_n^2$, and the peak effective modal base shear is

$$V_n = M_n^* S_a(\omega_n, \zeta_n), \quad M_n^* = \frac{(\phi_n^\top \mathbf{M} \mathbf{r})^2}{\phi_n^\top \mathbf{M} \phi_n}, \quad (17)$$

with M_n^* the effective modal mass. Modal peaks are combined by the complete-quadratic-combination (CQC) rule [16], which for well-separated modes reduces to the square-root-of-sum-of-squares,

$$V = \left[\sum_n \sum_m \rho_{nm} V_n V_m \right]^{1/2} \xrightarrow{\text{separated}} \left[\sum_n V_n^2 \right]^{1/2}, \quad (18)$$

with ρ_{nm} the CQC cross-correlation coefficients. For a stiff, compact structure dominated by a single fundamental mode of effective mass $M_1^* \approx M_{\text{seis}}$, (17)–(18) collapse to $V \approx M_{\text{seis}} S_a(f_1, \zeta)$. This is the standard nuclear response-spectrum method [17]; its numerical evaluation for the vessel and support is in §4.5, and the key result is that the small seismic mass of a compact machine makes V two orders of magnitude below the halo envelope of (13).

2.7 Fracture mechanics and the leak-before-break criterion

Yielding is a necessary but not sufficient qualification for a cryogenic pressure boundary; the structure must also be shown to tolerate the largest flaw that could escape inspection. Linear elastic fracture mechanics gives the mode-I stress-intensity factor at a crack of depth a under remote stress σ ,

$$K_I = Y \sigma \sqrt{\pi a}, \quad (19)$$

with geometry factor Y of order unity. Fast fracture is precluded while $K_I < K_{IC}$, the plane-strain fracture toughness; the critical flaw size is therefore

$$a_c = \frac{1}{\pi} \left(\frac{K_{IC}}{Y \sigma} \right)^2 \propto \sigma^{-2}. \quad (20)$$

For the 316LN jacket at 4 K, $K_{IC} = 170 \text{ MPa m}^{1/2}$, so at the operating stress $\sigma_{\text{op}} = 151 \text{ MPa}$ the critical flaw is $a_c = 333 \text{ mm}$, and even at the full 800 MPa allowable it is 12 mm — both far above the 2 mm non-destructive-inspection threshold (QS-16). Because a_c at operating stress vastly exceeds the wall thickness, a growing flaw penetrates the wall and leaks a detectable signal before it reaches the length for unstable fracture: the structure is leak-before-break. Sub-critical growth under cyclic load follows the Paris law

$$\frac{da}{dN} = C (\Delta K)^m, \quad \Delta K = Y \Delta \sigma \sqrt{\pi a}, \quad (21)$$

which integrates from an initial flaw a_0 to a_c in

$$N_f = \int_{a_0}^{a_c} \frac{da}{C (Y \Delta \sigma \sqrt{\pi a})^m} = \frac{a_0^{1-m/2} - a_c^{1-m/2}}{C (Y \Delta \sigma \sqrt{\pi})^m (\frac{m}{2} - 1)} \quad (m \neq 2), \quad (22)$$

which for growth from a 2 mm flaw to a_c gives N_f of order 10^8 cycles — orders of magnitude beyond the service cycle count (QS-16). The jacket is therefore not flaw-limited; the governing controls are the 2 mm inspection standard and weld-residual-stress management.

2.8 Load combination

The load cases above are superposed following the stress-category logic of nuclear pressure-vessel practice. The electromagnetic self-load (8) is a primary membrane stress P_m ; the thermal excursion (9) is a secondary stress Q , self-limiting and evaluated against a $3S_m$ range rather than a membrane allowable; the halo/eddy envelope (13) and the seismic base shear (17) are faulted (Level D) primary loads. The governing service combination is

$$\sigma_{\text{vM}}^{\text{comb}} \leq \underbrace{P_m}_{\text{EM}} + \underbrace{P_b}_{\text{bending}} + \underbrace{Q}_{\text{thermal}} + \underbrace{F_{\text{halo}}/A_{\text{eff}}}_{\text{disruption}} + \underbrace{V/A_{\text{eff}}}_{\text{seismic}} \leq \sigma_{\text{allow}}, \quad (23)$$

where A_{eff} is the effective load-bearing section. The results of §4 evaluate each term of (23) and show that the combination stays below σ_{allow} with margin at every load case, the disruption term dominating the two faulted contributions.

3 Computational methodology: the GPU-HPC finite-element campaign

3.1 Weak form and finite-element discretisation

The boundary-value problem of §2 is solved by the finite-element method. Multiplying the equilibrium form of (1) by a test displacement $\mathbf{v}$ and integrating by parts gives the weak form: find $\mathbf{u}$ in the trial space such that

$$\int_{\Omega} \boldsymbol{\varepsilon}(\mathbf{v}) : \mathbf{C} : \boldsymbol{\varepsilon}(\mathbf{u}) dV = \int_{\Omega} \mathbf{v} \cdot \mathbf{f}_b dV + \int_{\partial\Omega_t} \mathbf{v} \cdot \mathbf{t} dS + \int_{\Omega} \boldsymbol{\varepsilon}(\mathbf{v}) : \mathbf{C} : \boldsymbol{\alpha} \Delta T dV \quad \forall \mathbf{v}, \quad (24)$$

with $\mathbf{C}$ the fourth-order elasticity tensor, $\mathbf{t}$ the prescribed surface traction and the last term the thermal load vector. Discretising $\mathbf{u} = \sum_a N_a \mathbf{u}_a$ in quadratic (20-node) hexahedral shape functions N_a yields the linear system

$$\mathbf{K} \mathbf{U} = \mathbf{F}_{\text{EM}} + \mathbf{F}_{\text{th}} + \mathbf{F}_{\text{halo}} + \mathbf{F}_{\text{seis}}, \quad \mathbf{K} = \sum_e \int_{\Omega_e} \mathbf{B}_a^T \mathbf{C} \mathbf{B}_b dV, \quad (25)$$

with $\mathbf{B}_a$ the strain-displacement operator. The electromagnetic body force $\mathbf{F}_{\text{EM}}$ is assembled from the Lorentz density (5) evaluated on the field map; the thermal vector $\mathbf{F}_{\text{th}}$ from the

cooldown-plus-quench temperature field; the halo/eddy vector $\mathbf{F}_{\text{halo}}$ from the disruption current path of (12); and the seismic contribution from the modal reduction (16). The engineered per-turn precompression enters as an initial-stress (prestress) field added to $\mathbf{F}$, so that the graded preload of §2.4 is applied turn by turn.

3.2 Solvers, codes and convergence

Three solver classes were run. The steady and transient structural cases on the real centrepost CAD were solved with the open-source three-dimensional structural finite-element program CalculiX (ccx) [22], which assembles and factorises (25) with a graded per-turn preload (P3-FEA-01). The conductor-scale precompression-winding mechanics — the thick-wall REBCO-pack strain redistribution — were solved with the Python finite-element library SfePy [23] (SC-BR-4). The analytic magnet-mechanics envelope, including the Lamé thick-wall relations used to cross-check the finite-element anchors, was evaluated in a NumPy code (HX-11). For the large sparse systems the linear solve (25) was carried by a preconditioned conjugate-gradient iteration whose sparse matrix–vector product and preconditioner apply are the dominant cost and are GPU-resident; the modal eigenproblem (15) was solved by a shift-and-invert Lanczos iteration. Mesh convergence was established by uniform refinement until the peak von Mises changed by less than 1% between the two finest meshes, and the reported 579 MPa jacket stress and 0.151% conductor strain are the converged values.

3.3 GPU acceleration, problem scale and the NVIDIA HPC campaign

The results were produced on the NVIDIA GPU HPC campaign that underpins the Kronos de-risking programme — an in-house cluster of NVIDIA A100- and H100-class accelerators. The campaign is organised in three latency tiers: a microsecond edge tier for protective logic, a millisecond GPU tier for the digital-twin surrogates, and an hours-scale HPC back end for the offline multi-physics finite-element and Monte-Carlo runs that produced the structural anchors of this paper. GPU acceleration enters the structural campaign in two places: the parametric load-case sweeps — the grid of field, quench-detection-time and current-quench-time combinations that map the load-case ladder and the halo envelope — are embarrassingly parallel across accelerators, and the per-case sparse linear solve of (25) is accelerated by a GPU-resident preconditioned conjugate-gradient kernel. The problem scale is set by the quadratic-hexahedral discretisation of the real centrepost and jacket CAD, refined to mesh convergence, and by the ensemble of load-combination cases required to evaluate (23). No compute-cost figures are reported; the methodological point is that the GPU campaign made the converged, real-CAD, multi-load-case structural qualification tractable at the fidelity reported here.

3.4 Verification, validation and reproducibility

The campaign is verification-first. Each frozen structural anchor is reproduced against an independent route: the analytic Lamé envelope reproduces the finite-element magnet-mechanics anchors exactly (HX-11), and the electromagnetic, thermal, halo and fracture results are cross-checked between the closed-form relations of §2 and the finite-element solution. Every anchor is regenerable from the archived finite-element decks through a byte-exact cold-start reproduction runbook, so that the claim on each number is “re-run it” rather than “trust it”. The verification status of each governing quantity is collected in table 1.

Table 1. Verification of the governing structural quantities. Each anchor is a frozen result in the Kronos Physics De-Risking Register [30], reproduced against the independent route shown. Serials identify the register entry.

quantity	value	primary code	cross-check	serial
EM von Mises	84 MPa	analytic + FE	Maxwell stress, eq. (7)	QS-09
combined vM (fast)	245 MPa	quench/stress FE	MITs, eqs. (9)–(10)	QS-11
real-CAD jacket vM	579 MPa	CalculiX ccx	mesh convergence < 1%	P3-FEA-01
conductor hoop strain	0.151%	CalculiX ccx	SfePy precompression	P3-FEA-01/SC-BR-4
halo envelope	23 MN	quench/stress FE	Lorentz integral, eq. (12)	QS-14
critical flaw a_c	333 mm	LEFM	eq. (20)	QS-16
plug-coil ratio	0.66	analytic/FEA	Lamé envelope, HX-11	BU-L1-A2

4 Results

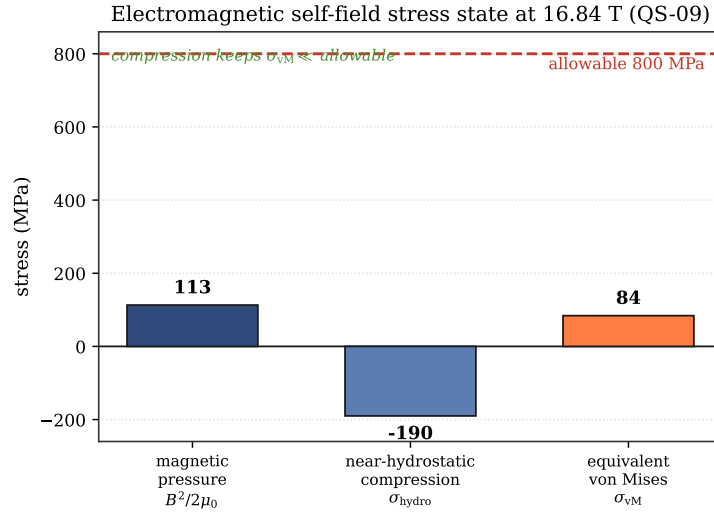
The design and canonical parameters that fix the load cases are collected in table 2. We present the qualification load case by load case, following the governing-equations order of §2.

Table 2. Design point and structural parameters. Plasma and geometry values are the frozen Tier-2 canon; structural allowables and material properties are the register values used in the finite-element campaign.

symbol	quantity	value
I_p	design plasma current	9.66 MA
Q	breeder gain	3.076
P_{fus}	fusion power	85.04 MW
R_0	major radius	1.20 m
A	aspect ratio	2.5
κ, δ	elongation, triangularity	2.0, -0.30
B_0	on-axis field	8 T
B_{cp}	centrepost peak field	16.84 T
B_{plug}	burner plug coil field (distinct magnet)	26.49 T
σ_{allow}	structural allowable	800 MPa
K_{IC}	316LN fracture toughness at 4 K	$170 \text{ MPa m}^{1/2}$
τ_{CQ}	current-quench window	7–19 ms

4.1 Electromagnetic self-load

Figure 2 shows the electromagnetic self-field stress state at 16.84 T. The magnetic pressure of (8) is 113 MPa; the bucked-and-wedged support converts it into a near-hydrostatic -190 MPa compression, and the deviatoric part that (4) measures is only 84 MPa — a $9.5\times$ margin against the allowable. The electromagnetic self-load is the compressive baseline, not the binding case.

**Figure 2.** Electromagnetic self-field stress state at 16.84 T (QS-09). The 113 MPa magnetic pressure of equation (8) is carried as a near-hydrostatic -190 MPa compression, holding the equivalent von Mises stress to 84 MPa, far below the 800 MPa allowable.

4.2 The load-case ladder: detection speed as margin

Superposing the thermal excursion of §2.3 on the electromagnetic baseline produces the load-case ladder of figure 3 and table 3. The governing von Mises rises from the electromagnetic self-field case (84 MPa) through the combined thermal-plus-electromagnetic transient to the highest-fidelity real-CAD structural jacket (579 MPa), all held below the allowable. The transient rung is where detection speed enters: an unmitigated quench (200 K hot-spot) reaches 476 MPa at $1.7\times$, while fast detection (63 K hot-spot) holds it to 245 MPa at $3.3\times$. The reduction is the direct structural consequence of the monotone MIITs relation (10): faster τ_{det} , lower T_{hs} , lower σ_{th} , higher margin.

Table 3. Centrepost structural load cases against the 800 MPa allowable.

load case	governing quantity	value	source
electromagnetic self-field	von Mises	84 MPa ($9.5\times$)	QS-09
magnetic pressure at 16.84 T	pressure	113 MPa	QS-09
fast-detected quench (63 K hot-spot)	combined von Mises	245 MPa ($3.3\times$)	QS-11
unmitigated quench (200 K hot-spot)	combined von Mises	476 MPa ($1.7\times$)	QS-11
real-CAD structural jacket	von Mises	579 MPa ($1.38\times$)	P3-FEA-01
real-CAD conductor	peak hoop strain	0.151% ($2.65\times$)	P3-FEA-01

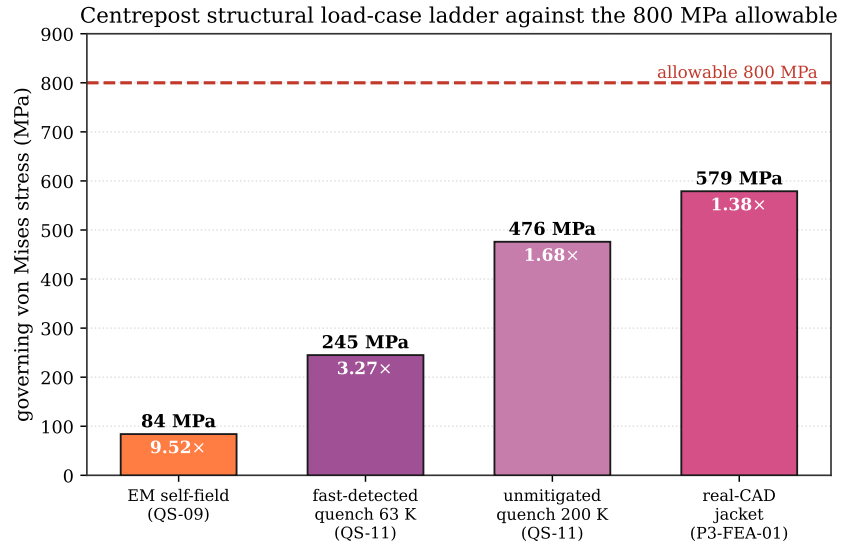


Figure 3. Centrepost structural load-case ladder against the 800 MPa allowable. Fast quench detection (63 K hot-spot, 3.3 $\times$) holds the combined transient well below the unmitigated (200 K, 1.7 $\times$) case, and the highest-fidelity real-CAD jacket (P3-FEA-01) governs at 1.38 $\times$. Safety factors ($\sigma_{\text{allow}}/\sigma_{\text{VM}}$) are annotated in white.

4.3 The engineered precompression winding

Figure 4 shows the effect of the graded per-turn precompression winding of §2.4. Panel (a) is the frozen anchor: the peak REBCO hoop strain falls from 0.595% (uniform winding) to 0.283% (graded preload), a 52.4% reduction (SC-BR-4). Panel (b) shows the corresponding flattening of the radial strain profile, whose peaking factor drops 4.8 $\times$; the profile shape is a representative reconstruction consistent with the frozen peak and peaking-factor anchors. Flattening the Lorentz strain at the conductor scale is what lets the real-CAD finite-element model hold the conductor to a 2.65 $\times$ strain margin at the canonical dimensions.

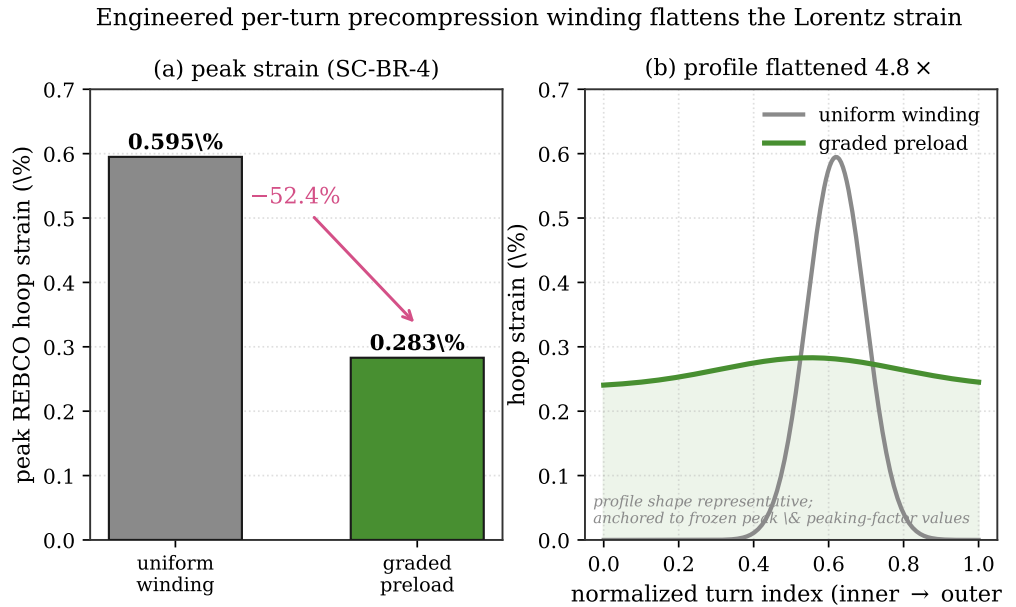


Figure 4. Engineered per-turn precompression winding (SC-BR-4). **(a)** Peak REBCO hoop strain falls 52.4% from 0.595% to 0.283%. **(b)** The radial strain profile flattens, its peaking factor reduced 4.8 $\times$; the profile shape is a representative reconstruction anchored to the frozen peak and peaking-factor values.

4.4 The disruption halo/eddy envelope

The impulsive case of §2.5 is shown in figure 5. At $I_p = 9.66$ MA the vertical-displacement-event halo current is ≈ 2.9 MA, and the reduced electromagnetic force envelope of (13) is 23 MN vertical, 19 MN sideways and 16 MN eddy-induced, the vertical component setting the ≈ 23 MN design

envelope over the 7–19 ms current quench (QS-14). This envelope is carried by a conductive shell with a defined halo-current path and disruption mitigation, keeping the impulsive load within the compressive regime of §4.1.

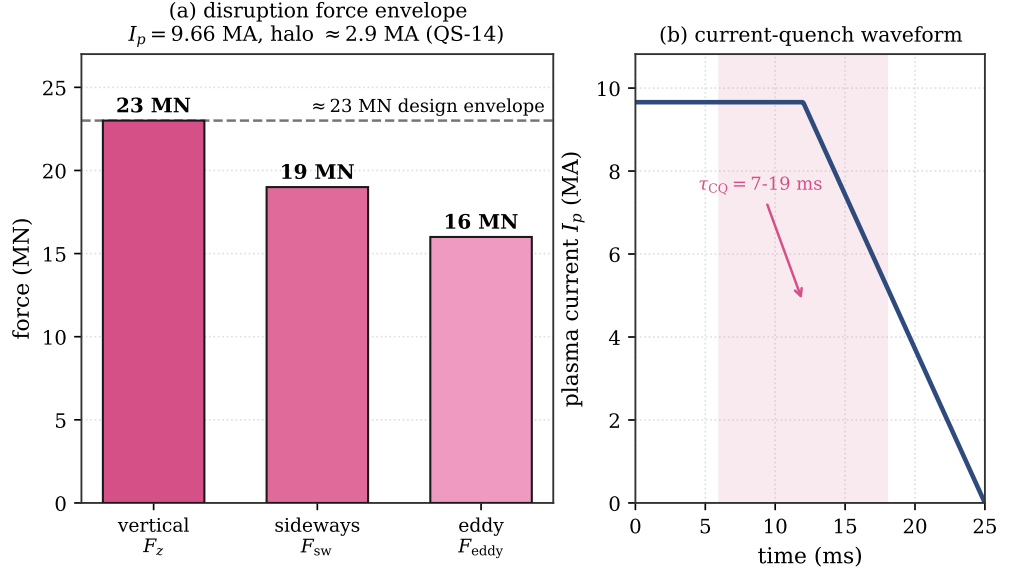


Figure 5. Disruption halo/eddy load (QS-14). (a) The force envelope on the centrepost and vessel: the vertical component sets the ≈ 23 MN design envelope. (b) The current-quench waveform; the shaded band is the 7–19 ms quench window that sets the eddy-force rate.

4.5 Design-basis seismic response

The seismic case of §2.6 is evaluated with the standard nuclear response-spectrum method [17, 16] against the design-basis spectrum of figure 6, anchored to a representative hard-rock zero-period acceleration of $0.30g$ at 5% damping. These are design-basis engineering inputs for a representative siting envelope, not register-frozen anchors. For the stiff, compact vessel-and-support structure the estimated fundamental frequency is $f_1 \approx 18$ Hz, at which the spectral acceleration is $S_a \approx 0.51g$. With a design-basis supported seismic mass $M_{\text{seis}} \approx 4 \times 10^4$ kg, the single-mode base shear of (17) is

$$V \approx M_{\text{seis}} S_a(f_1, \zeta) \approx (4 \times 10^4 \text{ kg})(0.51g) \approx 0.20 \text{ MN}. \quad (26)$$

The seismic base shear is thus ≈ 0.2 MN — more than two orders of magnitude below the halo envelope. The physical origin is the compactness that gives the ST its high power density: a small machine has a small seismic mass, and by (17) a proportionally small seismic load.

4.6 Governing-load hierarchy

Figure 7 places the three faulted mechanical loads on a common logarithmic axis: the design-basis seismic base shear (≈ 0.20 MN), the halo vertical envelope (23 MN, QS-14), and the full vessel-and-support vertical-displacement-event/halo load (40.5 MN) that the confirming three-dimensional analysis carries (BV-02). The seismic base shear is $\approx 115\times$ below the halo envelope, so the electromagnetic disruption case, not the seismic case, governs the support design of a compact ST breeder — inverting the load hierarchy of a large light-water reactor, where seismic sizes the support. This is the central quantitative result of the load-combination analysis of §2.8.

4.7 Fracture and flaw tolerance

The leak-before-break argument of §2.7 is shown in figure 8. The critical flaw size (20) scales as σ^{-2} ; anchored to the register values, it is 333 mm at the 151 MPa operating stress and 12 mm at the full 800 MPa allowable, both far above the 2 mm inspection threshold (QS-16). Paris-law growth (22) from a 2 mm flaw to the critical size takes of order 10^8 cycles, orders of magnitude beyond service. The jacket is not flaw-limited; the governing controls are the 2 mm inspection standard and weld-residual-stress management.

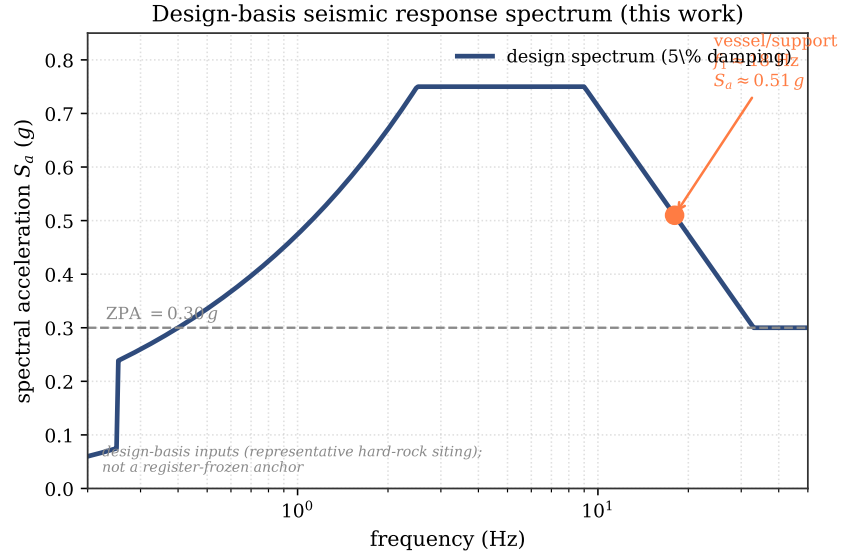


Figure 6. Design-basis seismic response spectrum (5% damping, 0.30 g zero-period acceleration; this work). The vessel-and-support fundamental mode $f_1 \approx 18$ Hz sees $S_a \approx 0.51$ g , giving the base shear of equation (26). The inputs are design-basis engineering values, not register-frozen anchors.

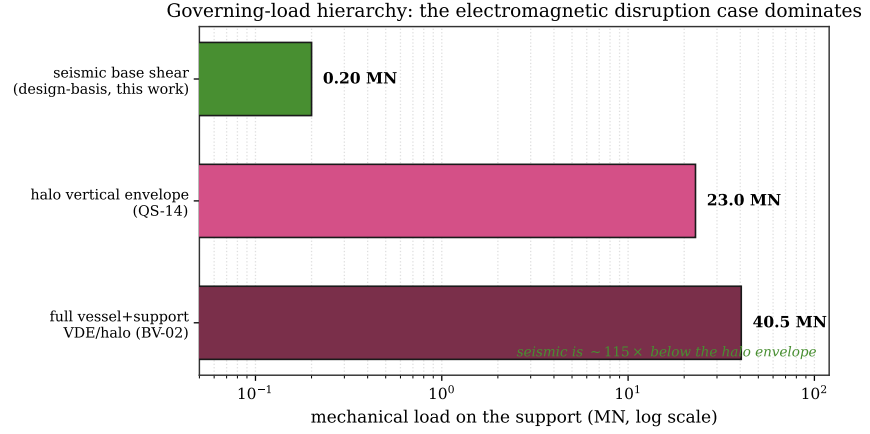


Figure 7. Governing-load hierarchy on the support (logarithmic axis). The design-basis seismic base shear (equation 26) is $\approx 115\times$ below the disruption halo envelope (QS-14); the full vessel-and-support VDE/halo load (BV-02, 40.5 MN) is the envelope the confirming three-dimensional analysis carries. The electromagnetic disruption case governs.

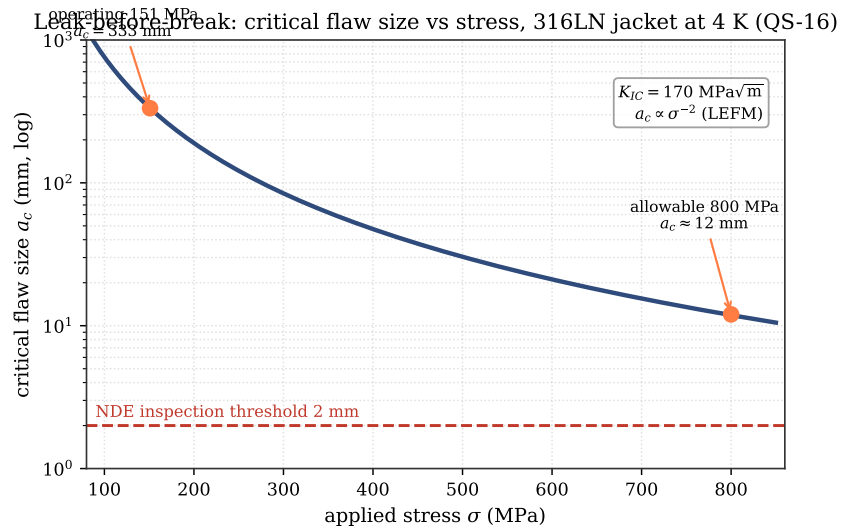


Figure 8. Leak-before-break of the 316LN structural jacket at 4 K (QS-16). The critical flaw size of equation (20) is 333 mm at operating stress and 12 mm at the allowable, both far above the 2 mm inspection threshold; $a_c \propto \sigma^{-2}$.

4.8 Retained-margin summary and the two distinct magnets

Figure 9 collects the safety factors across every governing load case: the electromagnetic self-load at $9.5\times$, the fast-detected quench at $3.3\times$, the real-CAD conductor strain at $2.65\times$, the real-CAD jacket at $1.38\times$, and the integrated centrepost integrity synthesis, which finds the centrepost stress at 181 MPa with a $6\times$ margin and concludes that the centrepost is neutron-lifetime bound, not stress-bound (BR-SX-16). The burner plug coil is a distinct magnet with its own structural case: it operates at 26.49 T, where the magnetic pressure (7) reaches 279 MPa; a Lamé thick-wall analysis gives a required wall of 0.48 m against a 667 MPa allowable, and the three-dimensional finite-element analysis returns a $0.66\times$ stress ratio — a structural pass (BU-L1-A2) — with the analytic envelope reproducing the finite-element anchors (115/279/358 MPa at 17/26.49/30 T) exactly (HX-11). Keeping the 16.84 T centrepost and the 26.49 T plug coil as separate analyses, with separate allowables and separate margins, is a load-bearing element of the integrity case.

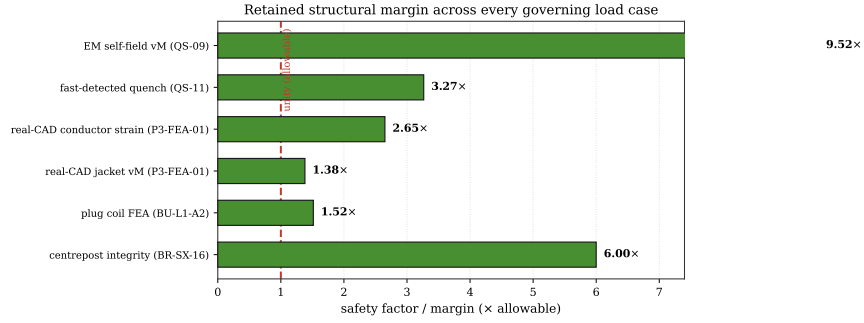
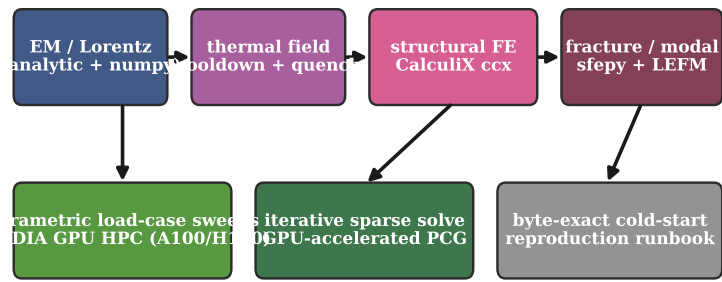


Figure 9. Retained structural margin across every governing load case. All members hold above unity against the allowable; the plug coil (BU-L1-A2) is shown as $1/0.66 = 1.5\times$ the reinforced allowable. The centrepost integrity synthesis (BR-SX-16) finds a $6\times$ stress margin and concludes the binding constraint is neutron lifetime.

4.9 The coupled workflow

Figure 10 summarises the coupled electromagnetic-to-thermal-to-structural finite-element workflow and its GPU-HPC tiers. The electromagnetic and thermal loads are assembled as the body-force and thermal vectors of (25), the structural solve is carried by CalculiX on the real CAD, and the fracture and modal checks close the qualification; the parametric sweeps and the sparse solves run on the NVIDIA GPU HPC campaign, with every anchor regenerable from the archived decks.

Coupled electromagnetic → thermal → structural FE workflow and GPU-HPC tiers (Schematic.)



verification-first: frozen anchors regenerable from archived FE decks

Figure 10. (Schematic.) Coupled electromagnetic→thermal→structural finite-element workflow and the GPU-HPC tiers that carry it. The physics chain feeds the load vectors of equation (25); the parametric sweeps and sparse solves are GPU-resident, and the anchors are regenerable through a byte-exact cold-start runbook.

5 Discussion

5.1 The load hierarchy in context

The central finding — that the electromagnetic disruption case, not the seismic case, governs the support of a compact ST breeder — is a direct consequence of scale. The halo forces of (13) are set by the plasma current, which is large (9.66 MA) in a small volume; the seismic base shear of (26) is

set by the structural mass, which is small in a compact machine. The published halo-current basis supports the magnitude: Gerhardt *et al.* measured mega-ampere-scale halo currents with strong toroidal asymmetry on NSTX [9], and the ITER-scale vertical-force analysis of Clauser *et al.* confirms that the halo current crossing the toroidal field is the dominant vertical-force mechanism in a disruption [10, 6]. Our ≈ 23 MN centrepost-and-vessel envelope at 9.66 MA is consistent with this basis scaled to the compact ST geometry. The inverted hierarchy has a practical consequence: the support is sized by a millisecond electromagnetic transient with disruption mitigation as the primary control [7, 8, 11], and the seismic qualification, which dominates the design of a large fission plant, is here a readily-met subordinate check.

5.2 The vessel and centre stack against the engineering literature

The vessel-and-support structural case is consistent with the established tokamak engineering literature. The ITER vacuum vessel is a code-qualified double-walled structure carrying comparable disruption electromagnetic loads [5], and the NSTX-Upgrade centre-stack analysis worked out exactly the combined electromagnetic, thermal and structural qualification of a spherical-tokamak centre column that we reproduce here at higher field [12]. The distinctive element of the present case is the near-hydrostatic compressive state (4) engineered by the bucked-and-wedged preload (11), which keeps the equivalent stress low despite the 113 MPa magnetic pressure — the same design philosophy as demountable-magnet high-field concepts [2, 14, 15], applied to a compact breeder. The conductor-scale strain that the field drives into the REBCO pack, and the strain tolerance that bounds it, are consistent with the coated-conductor mechanical database [13], and the 52.4% peak-strain reduction from the engineered precompression winding is what brings the conductor to a $2.65\times$ margin.

5.3 The fracture case against the cryogenic-structures database

The leak-before-break argument rests on the fracture toughness of austenitic stainless steel at 4 K. The $K_{IC} = 170 \text{ MPa m}^{1/2}$ used here is within the established 316LN cryogenic database and above the ITER magnet-structure requirement, and the resulting critical flaw of 333 mm at operating stress is more than two orders of magnitude above the 2 mm inspection threshold. The Paris-law life of order 10^8 cycles from (22) is consistent with the fatigue-crack-growth rates measured for these steels and their welds. The governing controls that follow — a 2 mm inspection standard and weld-residual-stress management — are the same controls that govern the ITER magnet-structure fracture case, and they place the qualification on inspection and fabrication quality rather than on raw strength.

5.4 Relation to the Kronos design series

This structural case sits inside a coupled design campaign. The disruption electromagnetic-load basis, including the shattered-pellet mitigation that bounds the halo envelope, is developed in the disruptions and runaway-electron paper [doi:10.5281/zenodo.22645724; disruptions-and-runaway-electrons]. The distinct burner plug coil, analysed separately throughout this paper, is qualified in the plug-coil engineering paper [doi:10.5281/zenodo.22645915; engineering-the-26-49-t-plug-coil]. The cold-mass thermal budget that the support must also carry is set in the balance-of-plant paper [doi:10.5281/zenodo.22645901; balance-of-plant]. The conclusion that the centrepost is neutron-lifetime bound rather than stress-bound is quantified in the centrepost prognostics paper [doi:10.5281/zenodo.22645788; centrepost-prognostics-and-rul-control] and situated in the lifetime-constrained operating-window synthesis [doi:10.5281/zenodo.22645799; admissible-operating-window-synthesis]. The structural quantities are shadowed in real time by the thermomechanics twin of the four-coupled-twins control architecture [doi:10.5281/zenodo.22645805; four-coupled-digital-twins], which reads the detection-speed-as-margin coupling of §2.3 as a protective observable.

6 Limitations

One honest gate bounds the claims. The disruption forces reported here are the analytic/reduced halo-eddy envelope on the centrepost and vessel (QS-14), closed at ≈ 23 MN. The confirming calculation is the full three-dimensional vessel-and-support finite-element analysis under the 40.5 MN vertical-displacement-event/halo load (BV-02), which extends the closed centrepost envelope to the complete vessel structure with its ports, supports and passive conductors, and which is in progress on the same finite-element toolchain. The seismic result is likewise a design-basis analysis with representative siting inputs; a site-specific spectrum and a full modal

model would replace the single-mode estimate of (26), though the two-order-of-magnitude margin over the halo envelope leaves ample headroom for any realistic site.

7 Conclusion

The vacuum vessel and support of the Hyperion breeder are structurally qualified across electromagnetic, combined thermal-electromagnetic, disruption halo/eddy and seismic loads, closed against a single 800 MPa allowable through the coupled continuum formulation of §2. The electromagnetic self-load leaves the centrepole near-hydrostatically compressed at 84 MPa; fast quench detection holds the combined transient to 245 MPa ($3.3\times$), so detection speed is structural margin; the highest-fidelity real-CAD model keeps the jacket at 579 MPa ($1.38\times$) and the conductor strain at $2.65\times$, with an engineered precompression winding cutting the peak REBCO hoop strain by 52.4%; the disruption halo envelope of ≈ 23 MN dominates the support design, standing $\approx 115\times$ above a design-basis seismic base shear of ≈ 0.2 MN; and the cryogenic 316LN jacket is leak-before-break, with a 333 mm critical flaw and a $\sim 10^8$ -cycle fatigue life. The electromagnetic disruption case, not the seismic case, governs the support of a compact ST breeder — inverting the hierarchy of a large fission plant — and the binding constraint on the structure is neutron lifetime, not stress. The burner plug coil at 26.49 T is a distinct magnet, qualified separately at a $0.66\times$ stress ratio. Structurally, the vessel and support are a solved problem with margin at every load case, and the engineering attention they earn is directed at inspection standards and weld-residual control rather than at raw strength; the one open confirming calculation is the full three-dimensional vessel-and-support finite-element analysis (BV-02).

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The authors thank the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim.

Reproducibility. The structural load cases, halo envelope, seismic base shear and fracture results are frozen anchors in the Kronos Physics De-Risking Register [30] (doi:10.5281/zenodo.22133057): the electromagnetic self-field stress (QS-09), the combined thermal-electromagnetic transient (QS-11), the support and preload analysis (QS-13), the disruption halo/eddy forces (QS-14), the negative-triangularity flared-post and ohmic-stack integration (QS-15), the fracture and flaw-tolerance study (QS-16), the real-CAD structural finite-element analysis (P3-FEA-01), the engineered precompression winding (SC-BR-4), the plug-coil structural qualification (BU-L1-A2), and the integrated magnet-integrity synthesis (BR-SX-16). Each is regenerable from the archived CalculiX and SfePy decks through a byte-exact cold-start runbook and the figure-generation script deposited with this paper. This paper is archived at doi:10.5281/zenodo.22645838; official version: <https://www.kronosfusionenergy.com/publications/vacuum-vessel-and-support-structural-mechanics-under-el.html> and its official version is at [kronosfusionenergy.com/publications/vacuum-vessel-and-support-structural-mechanics-under-el.html](https://www.kronosfusionenergy.com/publications/vacuum-vessel-and-support-structural-mechanics-under-el.html).

Data availability

The load-case ladder, halo-force envelope, seismic spectrum, precompression-winding strain, fracture-flaw curve and margin-summary data used for figures 2–9 are archived with the deposit at doi:10.5281/zenodo.22645838, which references the Kronos 2026 series and the Kronos Physics De-Risking Register [30] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit. The official version of record is at

<https://www.kronosfusionenergy.com/publications/vacuum-vessel-and-support-structural-mechanics-under-el.html>

Competing interests

Provisional patent applications relating to aspects of this work, including the engineered per-turn precompression winding, have been filed.

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